

Neural Network Classification of Alzheimer's Disease from Structural Brain MRI: A Patient-Level Evaluation of Convolutional and Transfer Learning Approaches

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Abstract

Alzheimer's disease (AD) is the leading cause of dementia worldwide, and early detection from structural MRI remains an active research area. This study evaluated whether convolutional neural networks (CNNs) can classify AD from axial brain MRI slices, and whether transfer learning with EfficientNetB0 (pretrained on ImageNet) improves on training from scratch with a small medical imaging dataset. Data were drawn from OASIS-1 and OASIS-2, restricted to patients aged 60 and older and labeled non-demented (CDR = 0) or Alzheimer's (CDR > 0). After skull stripping, registration of both cohorts to a common template, and extraction of 10 axial slices per patient, 318 patients (3,180 images) were split at the patient level into training (n = 222), validation (n = 48), and test (n = 48) sets, stratified by label and source. Three models were compared against a majority-class baseline: a CNN trained from scratch, the same CNN with light augmentation and early stopping, and a two-phase EfficientNetB0 transfer learning model. On the test set, the augmented CNN reached 77.1% patient-level accuracy (F1 = 0.77, AUC = 0.80), outperforming both the baseline (52.1%) and transfer learning (58.3%, AUC = 0.67). These results indicate that, at this sample size, a compact CNN trained directly on the target domain generalizes better than ImageNet-pretrained features, and underscore the importance of patient-level data splitting for producing trustworthy performance estimates in this literature.

Introduction

Alzheimer's disease is a progressive neurodegenerative disorder and the most common cause of dementia. An estimated 57 million people worldwide were living with dementia in 2019, a figure projected to nearly triple by 2050 (GBD 2019 Dementia Forecasting Collaborators, 2022). AD currently has no cure, but earlier detection allows earlier intervention and care planning. Structural MRI shows well-established atrophy markers, chiefly enlargement of the lateral ventricles, and disease severity is clinically staged using the Clinical Dementia Rating (CDR), a clinician-administered scale independent of imaging (Morris, 1993).

CNNs can, in principle, learn atrophy patterns directly from pixel data, but medical imaging datasets are typically far smaller than the natural-image corpora (e.g., ImageNet; Deng et al., 2009) used to develop modern computer vision methods, which has motivated transfer learning from pretrained backbones such as EfficientNetB0 (Tan & Le, 2019). A separate, largely methodological concern affects much published work on these same OASIS datasets: data leakage. When slices from the same patient are distributed across training and test sets, a model can effectively memorize and later "recognize" that patient, inflating accuracy without reflecting true generalization (Kapoor & Narayanan, 2023). This study addressed three questions: (a) can a CNN trained on brain MRI slices distinguish AD from non-demented status under a leakage-free, patient-level split; (b) does EfficientNetB0 transfer learning improve on a small CNN trained from scratch at this sample size; and (c) what performance can be achieved by combining two independently acquired cohorts, OASIS-1 and OASIS-2?

Methodology

Design and data. This retrospective comparative study used two releases of the Open Access Series of Imaging Studies (OASIS): OASIS-1, cross-sectional T1-weighted MRI already registered and skull-stripped (Marcus et al., 2007), and OASIS-2, a longitudinal collection distributed as raw scanner output (Marcus et al., 2010). Patients were restricted to age ≥ 60 and relabeled non_demented (CDR = 0) or alzheimer (CDR > 0). From OASIS-1, 198 patients met criteria (100 alzheimer, 98 non-demented). From OASIS-2, only each patient's first visit was kept (to prevent longitudinal leakage) and the "Converted" group was excluded (CDR assigned at conversion, not baseline), leaving 120 patients. The combined cohort comprised 318 patients and, at 10 axial slices per patient, 3,180 images.

Preprocessing. OASIS-1 volumes were already registered/skull-stripped; 10 axial slices (indices 80–120, capturing the lateral ventricles and temporal lobes) were extracted with nibabel and intensity-normalized (0–255).

OASIS-2 required a five-step pipeline built with ANTsPy/ANTsPyNet: axis reorientation; construction of a template by averaging 30 aligned OASIS-1 volumes; U-Net brain extraction constrained to a slice-by-slice convex hull (to avoid erroneously removing low-intensity structures such as the thalamus); rigid-then-affine registration to the template; and the identical slice-extraction step used for OASIS-1. Both template and moving volumes were built with `ants.from_numpy` so they shared a common origin/orientation, since building them differently caused registration to converge to a coronal, rather than axial, local minimum. Intensity checks confirmed both cohorts occupied a comparable domain before normalization.

Patient-level split. The 318 patients were split 70/15/15 into training (222), validation (48), and test (48) sets by stratified sampling on label and source, with all 10 slices from a given patient assigned to the same partition. This safeguard is central to the study: because a patient's 10 slices are visually near-identical, an image-level split would let a model "recognize" a test patient seen in training, inflating accuracy — a pattern observed when checked informally against several public analyses of these same datasets.

Models. (1) A scikit-learn majority-class DummyClassifier. (2) A 3-block CNN (16/32/64 filters, 3×3 , ReLU, max pooling; flatten; dense-64; softmax; 224×224 grayscale input; Adam; 25 epochs), trained without regularization. (3) The same CNN with light rotation (± 0.02 rad) and translation ($\pm 2\%$) augmentation — zoom was excluded because it directly rescales the ventricle-to-brain ratio that is the discriminative signal — and early stopping on validation accuracy (patience = 10; val_loss was unsuitable, as it minimized within epoch 1). (4) EfficientNetB0 transfer learning (ImageNet-pretrained; Deng et al., 2009; Tan & Le, 2019), Phase 1 with a frozen backbone and a new head (global average pooling, dense-128, softmax; 164,226 trainable parameters; 20 epochs, patience = 7), Phase 2 unfreezing the last 20 backbone layers (BatchNormalization layers kept frozen) at a $100 \times$ lower learning rate (20 epochs, patience = 8).

Evaluation. Slice-level predictions were aggregated per patient by averaging class probabilities across each patient's 10 slices (soft voting). Accuracy, precision, recall, F1, balanced accuracy, and AUC were computed at this patient level on held-out validation/test sets; test predictions were also cross-tabulated against each patient's original CDR score.

Tools. Python, TensorFlow/Keras, scikit-learn, ANTsPy/ANTsPyNet, Google Colab.

Results

On the test set (25 non-demented, 23 Alzheimer's patients), the majority-class baseline reached 52.1% accuracy (macro F1 = 0.34; balanced accuracy and AUC = 0.50 by construction). The from-scratch CNN reached training accuracy near 1.0 by epoch 15–20 while validation accuracy remained near 0.60–0.65; at the patient level on the validation set it reached 71% accuracy (non_demented: precision = 0.70, recall = 0.81; alzheimer: precision = 0.72, recall = 0.59).

The augmented CNN reached 77.1% test accuracy, balanced accuracy = 0.768, AUC = 0.798 (non_demented: precision = 0.75, recall = 0.84; alzheimer: precision = 0.80, recall = 0.70; confusion matrix: 21 true negatives, 4 false positives, 7 false negatives, 16 true positives). Cross-tabulated against CDR, specificity was 0.84 for CDR = 0 ($n = 25$), recall was 0.67 for CDR = 0.5 ($n = 18$), and recall was 0.80 for CDR = 1.0 ($n = 5$).

EfficientNetB0 Phase 1 plateaued near 61–62% validation accuracy; Phase 2 produced only marginal further change. On the test set, the fine-tuned transfer model reached 58.3% accuracy, balanced accuracy = 0.570, AUC = 0.671 (non_demented: precision = 0.56, recall = 0.88; alzheimer: precision = 0.67, recall = 0.26; confusion matrix: 22 true negatives, 3 false positives, 17 false negatives, 6 true positives). Table 1 summarizes patient-level test-set performance across all models.

Table 1. Patient-Level Test-Set Performance by Model ($n = 48$)

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	AUC
Majority-class baseline	0.52	0.26	0.50	0.34	0.50
CNN, augmented + early stopping	0.77	0.78	0.77	0.77	0.80
Transfer learning (EfficientNetB0)	0.58	0.62	0.57	0.53	0.67

Note. All metrics computed at the patient level using soft-voted probabilities across each patient's 10 slices.

Discussion

A compact CNN trained directly on the target domain, with only light, physiologically appropriate augmentation and early stopping, outperformed ImageNet-pretrained EfficientNetB0 at this sample size (222 training patients). This is a documented pattern in low-data medical imaging: ImageNet features represent edges, textures, and shapes optimized for natural photographs, a mismatch for a discriminative signal that is largely geometric and domain-specific (ventricle-to-brain ratio), while fine-tuning a multi-million-parameter backbone on a few hundred images invites overfitting or collapse toward the majority class — consistent with Phase 2 fine-tuning yielding essentially no improvement over the frozen Phase 1 head.

The patient-level split was the central methodological safeguard: because a patient's 10 slices are visually near-identical, an image-level split would let a model "recognize" a previously seen patient, producing an inflated, non-generalizing accuracy estimate. Kapoor and Narayanan (2023) documented this exact failure mode across 294 papers in 17 scientific fields, finding it frequently erased the claimed advantage of complex models over simple baselines. The comparatively modest accuracies reported here (58–77%), well below the near-ceiling accuracies sometimes reported for automated OASIS classification, should be read as a more defensible estimate of generalization to new patients rather than evidence of weaker modeling.

Excluding zoom from augmentation, while retaining rotation/translation, reflects a broader principle: an augmentation is valid only if it preserves the label-relevant signal, and zoom directly rescales the ventricular ratio that defines the positive class here. The severity-stratified pattern (lower recall at CDR = 0.5 than at CDR = 1.0) fits an expected clinical difficulty gradient, since very early atrophy is harder to distinguish from normal aging for both automated systems and human raters — pointing to sensitivity at the very-mild stage as the most clinically valuable, and most difficult, target for future improvement.

Several factors should temper these conclusions and motivate future work: the training set remains modest (222 patients) even after combining cohorts; each scan contributed only 10 two-dimensional slices rather than the full volume, which 2.5-D or 3-D architectures could exploit; a backbone pretrained on radiological rather than natural images might provide a more relevant prior than EfficientNetB0; and evaluation relied on a single 48-patient test split rather than cross-validation, so point estimates carry non-trivial sampling uncertainty.

Conclusion

Using a leakage-safe, patient-level evaluation, a compact CNN trained from scratch with light, physiologically appropriate augmentation and early stopping reached 77.1% test accuracy ($F1 = 0.77$, $AUC = 0.80$) classifying Alzheimer's disease from brain MRI in a combined OASIS-1/OASIS-2 cohort of 318 patients, outperforming both a majority-class baseline (52.1%) and EfficientNetB0 transfer learning (58.3%, $AUC = 0.67$). These findings support two lessons for small-sample medical imaging: patient-level evaluation should be treated as a prerequisite, not a refinement, since it materially changes which conclusions are trustworthy; and transfer learning from natural-image datasets should not be assumed to help when the discriminative signal is domain-specific and geometric. Future work should prioritize larger combined cohorts, volumetric (3-D) architectures, and domain-relevant pretrained backbones.

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